

C-DAC Mumbai

Lab Assignment

Problem 1: Sum of Two Numbers (Using a Method)

```
import java.util.Scanner;

class sumof2{
    static int sumOfTwoNumbers(int a, int b){
        int result = a + b;
        return result;
    }
    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number : ");
        int a = sc.nextInt();

        System.out.print("Enter second number : ");
        int b = sc.nextInt();

        int sum = sumOfTwoNumbers(a,b);
        System.out.println("The sum of " + a + " and " + b + " is " + sum);
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java sumof2
Enter first number : 15
Enter second number : 25
The sum of 15 and 25 is 40

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java sumof2
Enter first number : 10
Enter second number : 15
The sum of 10 and 15 is 25

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>
```

Problem 2: Simple Age Checker (Using a Method)

```
import java.util.Scanner;

public class AgeChecker {
    public static void checkAgeCategory(int age){
        if(age < 18){
            System.out.println("Aww, you're still a kid!");
        }
    }
}
```

```

else if((age>=18) && (age<=59)){
    System.out.println("Congrats! You're officially an adult");
}
else if(age >= 60){
    System.out.println("Welcome to the senior club!");
}
else{
    System.out.println("Invalid Number ");
}
}

public static void main(String args[]){
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter Age : ");
    int age = sc.nextInt();
    checkAgeCategory(age);
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>javac AgeChecker.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java AgeChecker
Enter Age : 22
Congrats! You're officially an adult
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java AgeChecker
Enter Age : 30
Congrats! You're officially an adult
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>

```

Problem 3: Print Even Numbers (Using while Loop)

```

public class EvenNumber {
    public static void printEvenNumbers(){
        int n = 1;
        while(n <= 50){
            if(n % 2 == 0){
                System.out.print(" " + n);
            }
            n++;
        }
    }

    public static void main(String[] args) {
        printEvenNumbers();
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>javac EvenNumber.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java EvenNumber
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>

```

Problem 4: User Input for Positive Numbers (Using do-while Loop)

```
import java.util.Scanner;

public class Positivenumber {
    public static void askForPositiveNumber(){
        Scanner sc = new Scanner(System.in);
        do{
            System.out.print("Enter Number : ");
            int n = sc.nextInt();
            if(n > 0){
                System.out.println("You entered a positive number : " + n);
                break;
            }
        }while(true);
    }
    public static void main(String args[]){
        askForPositiveNumber();
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java Positivenumber
Enter Number : -5
Enter Number : 0
Enter Number : 8
You entered a positive number : 8

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java Positivenumber
Enter Number : -9
Enter Number : 6
You entered a positive number : 6

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>
```

Problem 5: Print Multiplication Table (Using for Loop)

```
import java.util.Scanner;

public class Multiplication {
    public static void printMultiplicationTable(int n){
        for(int i = 1; i <= 10 ; i++){
            System.out.println(n + " * " + i + " = " + n*i);
        }
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a Number : ");
        int n = sc.nextInt();
    }
}
```

```

        printMultiplicationTable(n);
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>javac Multiplication.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java Multiplication
Enter a Number : 7
7 * 1 = 7
7 * 2 = 14
7 * 3 = 21
7 * 4 = 28
7 * 5 = 35
7 * 6 = 42
7 * 7 = 49
7 * 8 = 56
7 * 9 = 63
7 * 10 = 70

```

Problem 6: Calculate the Sum of Numbers from 1 to N (Using for Loop)

```
import java.util.Scanner;
```

```

public class sum1ton {
    public static int calculateSum(int n){
        int sum = 0;
        for(int i = 1; i<= n; i++){
            sum = sum + i;
        }
        return sum;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Number : ");
        int n = sc.nextInt();
        System.out.println("The sum of numbers from 1 to " + n + " is : " + calculateSum(n));
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>javac sum1ton.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java sum1ton.java
Enter Number : 5
The sum of numbers from 1 to 5 is : 15

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java sum1ton.java
Enter Number : 10
The sum of numbers from 1 to 10 is : 55

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>|

```

Bonus Problem: Menu-Driven Java Program (Switch-Case)

```
import java.util.Scanner;
```

```
public class Bonus {  
    public static void main(String args[]){  
        Scanner sc = new Scanner(System.in);  
        int choice;  
  
        do {  
  
            System.out.println("Enter 1 - Grade Evaluation");  
            System.out.println("Enter 2 - Leap Year");  
            System.out.println("Enter 3 - Day of the Week");  
            System.out.println("Enter 4 - Default Values");  
            System.out.println("Enter 5 - Exit");  
            System.out.print("Enter your choice: ");  
            choice = sc.nextInt();  
  
            switch(choice){  
                case 1: {  
                    System.out.print("Enter Marks in Maths : ");  
                    int m = sc.nextInt();  
                    System.out.print("Enter Marks in Science : ");  
                    int s = sc.nextInt();  
                    System.out.print("Enter Marks in History : ");  
                    int h = sc.nextInt();  
  
                    int avg = (m + s + h)/3;  
                    System.out.println("Average Marks : " + avg);  
  
                    if( avg >= 90){  
                        System.out.print("Grade A");  
                    }  
                    else if( (avg >= 70) && (avg <= 89)){  
                        System.out.print("Grade B");  
                    }  
                    else if((avg >= 50) && (avg <= 69)){  
                        System.out.print("Grade C");  
                    }  
                    else if((avg >= 30) && (avg <= 49)){  
                        System.out.print("Grade D");  
                    }  
                    else{  
                        System.out.print("Fail");  
                    }  
                    break;  
                }  
            }  
        }  
    }  
}
```

```
case 2:{
System.out.print("Enter year : ");
int y = sc.nextInt();

if(((y % 4 == 0) && (y % 100 != 0)) || (y % 400 == 0)){
    System.out.print(y + " is a Leap Year ");
}
else{
    System.out.print(y + " is not a Leap Year ");
}
break;
}

case 3:{
System.out.print("Enter day number : ");
int day = sc.nextInt();

switch(day){
    case 1: System.out.print("Monday"); break;
    case 2: System.out.print("Tuesday"); break;
    case 3: System.out.print("Wednesday"); break;
    case 4: System.out.print("Thursday"); break;
    case 5: System.out.print("Friday"); break;
    case 6: System.out.print("Saturday"); break;
    case 7: System.out.print("Sunday"); break;
    default: System.out.print("Invalid day number"); break;
}
break;
}

case 4:{
byte a = 25;
short b = 100;
int c = 1000;
long d = 10000L;
float e = 10.5f;
double f = 20.55;
char g = 'A';
boolean h = true;

System.out.println(a);
System.out.println(b);
System.out.println(c);
System.out.println(d);
System.out.println(e);
```

```

        System.out.println(f);
        System.out.println(g);
        System.out.println(h);
        break;
    }

    case 5:{
        System.out.println("Exiting the program...");
        break;
    }

    default:{
        System.out.println("Invalid choice. Please enter 1 to 5.");
    }
}
System.out.println("\n*****");
} while(choice != 5);
}
}

```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>javac Bonus.java
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 3\Assignment>java Bonus
```

```
Enter 1 - Grade Evaluation
```

```
Enter 2 - Leap Year
```

```
Enter 3 - Day of the Week
```

```
Enter 4 - Default Values
```

```
Enter 5 - Exit
```

```
Enter your choice: 2
```

```
Enter year : 2004
```

```
2004 is a Leap Year
```

```
*****
```

```
Enter 1 - Grade Evaluation
```

```
Enter 2 - Leap Year
```

```
Enter 3 - Day of the Week
```

```
Enter 4 - Default Values
```

```
Enter 5 - Exit
```

```
Enter your choice: 3
```

```
Enter day number : 1
```

```
Monday
```

```
*****
```

```
Enter 1 - Grade Evaluation
```

```
Enter 2 - Leap Year
```

```
Enter 3 - Day of the Week
```

```
Enter 4 - Default Values
```

```
Enter 5 - Exit
```

```
Enter your choice: 1
```

```
Enter Marks in Maths : 95
```

```
Enter Marks in Science : 90
```

```
Enter Marks in History : 99
```

```
Average Marks : 94
```

```
Grade A
```

```
*****
```

```
Enter 1 - Grade Evaluation
```

```
Enter 2 - Leap Year
```

```
Enter 3 - Day of the Week
```

```
Enter 4 - Default Values
```

```
Enter 5 - Exit
```

```
Enter your choice: 5
```

```
Exiting the program...
```

```
*****
```